MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, THYRATRON

## TYPE 1258

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen

See figure 1

Mounting position: Any

Weight: 0.5-ounce nominal (14.18 grams)

ABSOLUTE RATINGS:

| Parameter: | Ef | epy | epx | Ebb | Ecc | egx | egy | ib |
|---|---|---|---|---|---|---|---|---|
| Unit: | Vac | kv | kv | Vdc | Vdc | v | v | a |
| Maximum: | --- | 1.0 | 1.0 | --- | --- | 200 | (Note 4) | 20 |
| | (Note 1) | (Note 2) | (Note 3) | | | | | |
| Minimum: | --- | --- | --- | 300 | --- | --- | --- | --- |
| TEST CONDITIONS: | 6.3 | 1.0 | --- | --- | 0 | --- | 130 | --- |

ABSOLUTE RATINGS:

| Parameter: | Ib | tk | $\frac{dik}{dt}$ | Pb | Ehk | prr | TA |
|---|---|---|---|---|---|---|---|
| Unit: | mAdc | sec | a/µs | --- | V | --- | °C |
| Maximum: | 50 | --- | 400 | $0.1 \times 10^9$ | +25 | --- | +90 |
| Minimum: | --- | 60 | --- | --- | -100 | --- | -50 |
| TEST CONDITIONS: | --- | 60 | --- | --- | --- | 5,000 (See figure 2) | --- |

GENERAL:

Qualification - Required

Ⓖ denotes changes



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | | | | |
| --- | Direct current holdoff | 10 | egy = 0; Rp = Rg = 1,000 ohms; tk = 20 | 0.65 | II | Ebb | 1,300 | --- | Vdc |
| 3241 | Heater current | - | | 0.65 | II | If | 1.6 | 2.0 | Aac |
| 3267 | Instantaneous starting | 5,11 | epy = 1,000 v (min) | 0.65 | II | --- | --- | --- | --- |
| (G)3247 | DC anode voltage for conduction | - | | 0.65 | II | Ebb | --- | 100 | Vdc |
| 3246 | Operation | 5,6, 7 | epy = 1.3 kv (min) | 0.65 | II | egy | --- | 130 | v |
| 3251 | Pulse emission (method A) | - | ik = 20 a (min); prr = 60 ±10%; tp = 5.0 µs ±10%; tr = 0.5 µs (max); specified time interval = 2.5 µs | 0.65 | II | egk | --- | 200 | v |
| | Quality conformance inspection, part 2 | | | | | | | | |
| 1031 | Low-frequency vibration | 12 | DC holdoff test, except Ebb = 250 Vdc (min) | --- | --- | --- | --- | --- | --- |
| 1041 | Shock | - | No voltages; hammer angle = 48° | --- | --- | --- | --- | --- | --- |
| --- | Shock-test end points: | | | | | | | | |
| 3246 | Operation | 5,6 | | --- | --- | egy | --- | 130 | v |
| 3247 | DC anode voltage for conduction | - | | --- | --- | Ebb | --- | 100 | Vdc |
| 3261 | Time jitter | - | | --- | --- | tj | --- | 0.01 | µs |
| 1336 | Heater-cathode leakage | - | Ehk = -100 Vdc, +50 Vdc | --- | --- | Ihk | --- | 20 | µAdc |
| 3256 | Anode delay time | 5,6 | Operation; t = 120 | --- | --- | tad | --- | 0.6 | µs |
| 3256 | Anode delay time drift | 13 | Anode delay time | --- | --- | Δtad | --- | 0.15 | µs |
| 3261 | Time jitter | 5 | epy = 200 v (max) | --- | --- | tj | --- | 0.01 | µs |
| | Quality conformance inspection, part 3 | | | | | | | | |
| --- | Life test | 5 | Group B; egy = 140 v; t = 96 hours "on" and 1 hour "off" (tube mounted horizontally); t = 500 hours | --- | --- | --- | --- | --- | --- |
| --- | Life-test end points: | | | | | | | | |
| 3246 | Operation | 5,6 | | --- | --- | egy | --- | 140 | v |
| 3247 | DC anode voltage for conduction | - | | --- | --- | Ebb | --- | 250 | Vdc |
| 3261 | Time jitter | - | egy = 140 v | --- | --- | tj | --- | 0.02 | µs |
| 1031 | Vibration fatigue | 14 | No voltages | --- | --- | --- | --- | --- | --- |



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 3 -Continued | | | | | | | | |
| --- | Vibration-fatigue test end points: | | | | | | | | |
| 3246 | Operation | 5,6 | | --- | --- | egy | --- | 130 | v |
| 3247 | DC anode voltage for conduction | - | | --- | --- | Ebb | --- | 100 | Vdc |
| 3261 | Time jitter | - | | --- | --- | tj | --- | 0.01 | µs |
| 1031 | Vibration | 8,9, 14 | t = 240; Ef = 6.3 V; egy = 0; Rp = Rg = 1,000 ohms; Ehk = -100 V | --- | --- | Ebb | 1,300 | --- | Vdc |
| (G) 3246 | Operation at elevated temperature | 5,14 | t = 5.0 hours; TA = +100°C | --- | --- | egy | --- | 130 | v |

NOTES:

1. Cathode temperature is determined by rms cathode current as well as by heater power. The average heater voltage of 6.3 V therefore is applicable only near full operating conditions. At light loading it is recommended to operate the heater voltage higher. Recommended values for various operating conditions may be determined from figure 3. Figure 4 is for application information only. The heater voltage shall be held within +5 percent or -10 percent of the values shown on figure 3.

2. For instantaneous starting applications where anode voltage is applied instantaneously, the power-supply filter design shall be such that the maximum permissible epy is 1,000 v and shall not be attained in less than 0.04 second.

3. In pulsed operation, the peak inverse voltage shall not exceed 500 v during the first 25 µs after the pulse.

4. Driver pulse, measured at tube socket with thyratron grid disconnected: egy = 175 v (min), time of rise = 0.5 µs (max), grid pulse duration = 2.0 µs (min). Impedance of drive circuit = 1,000 ohms (max).

5. The circuit constants shall be chosen so that at epy = 1.0 kv under resonant charging conditions, dik/dt = 400 a/µs (min), ib = 20 a, tp = 0.50 µs ±10 percent, prr = 5,000. The grid pulse characteristics shall be tp = 2.0 µs (max), tr = 0.5 µs (min), driver impedance = 1,000 ohms (min).

6. The tube shall operate continuously for 5 minutes without evidence of continuous conduction, anode heating, or corona discharge.

7. This test is to be the first test performed at the conclusion of the holding period.

8. The tube under test shall be vibrated in position Y at a frequency of 10 to 500 to 10 Hz with acceleration of 2.5 G. Failure to holdoff dc anode voltage during the vibration period shall constitute a failure. Heater-cathode shorts of 100,000 ohms or less having a duration of 100 µs or more, shall be cause for rejection.

9. The dc holdoff voltage test shall be run concurrently with the vibration test.

10. This test shall be performed after the 96-hour holding period. After a heater warmup time of 60-seconds minimum, increase the dc anode voltage from 0 to 1,300 Vdc. Hold at 1,300 Vdc for a period of 30-seconds minimum. No breakdown shall occur during this period.

11. The tube shall operate satisfactorily on pushbutton starting within three attempts when anode voltage is applied in such a manner as to rise from 0 to 1,000 v within 0.03 second (the filter design shall be such that at least 500 v is reached within 0.015 second). Any tube failing to start within three attempts shall be considered a failure.

12. Heater-cathode shorts of 100,000 ohms or less, having a duration of 100 µs or more, shall be cause for rejection.

13. During the interval between 2 minutes and 7 minutes of the anode delay time test, the Δ tad relative to the tad value observed on the anode delay time test shall not exceed the specified value.

(G) 14. This test shall be performed annually. A sample size of three tubes shall be used, with no failures allowed. In case of a sample failure, the test shall become a quality conformance inspection (QCI), part 2 test, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3 test.



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Custodians:
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Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH

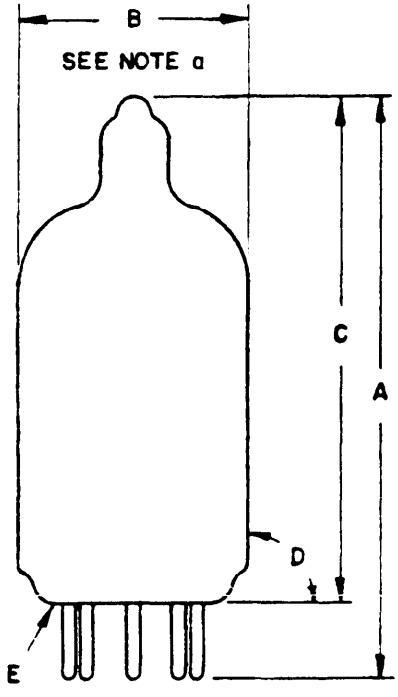
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2881)



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| PIN CONNECTIONS | |
|---|---|
| Pin No. | Element |
| 1 | g1 |
| 2 | k |
| 3 | h |
| 4 | h |
| 5 | int con |
| 6 | a |
| 7 | int con |

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 2 | | |
| A | --- | 2.125 (53.98) |
| B | .804 (20.42) | .844 (21.44) |
| C | --- | 1.875 (47.63) |
| D | 86 1 '2 | 93 1 '2 |
| Quality conformance inspection, part 3 (see note b) | | |
| E | Base: E7-1 (EIA) | |

NOTES:
a. Minimum measurement applies in area between bottom and top of bulb.
b A sample size of three tubes shall be checked annually, with no failures allowed. In case of a sample failure, the test shall become a quality conformance inspection (QCI), part 2 test, for three successful consecutive submissions at which time the test may revert to the QCI, part 3 test.

FIGURE 1. Outline drawing of electron tube type 1258.



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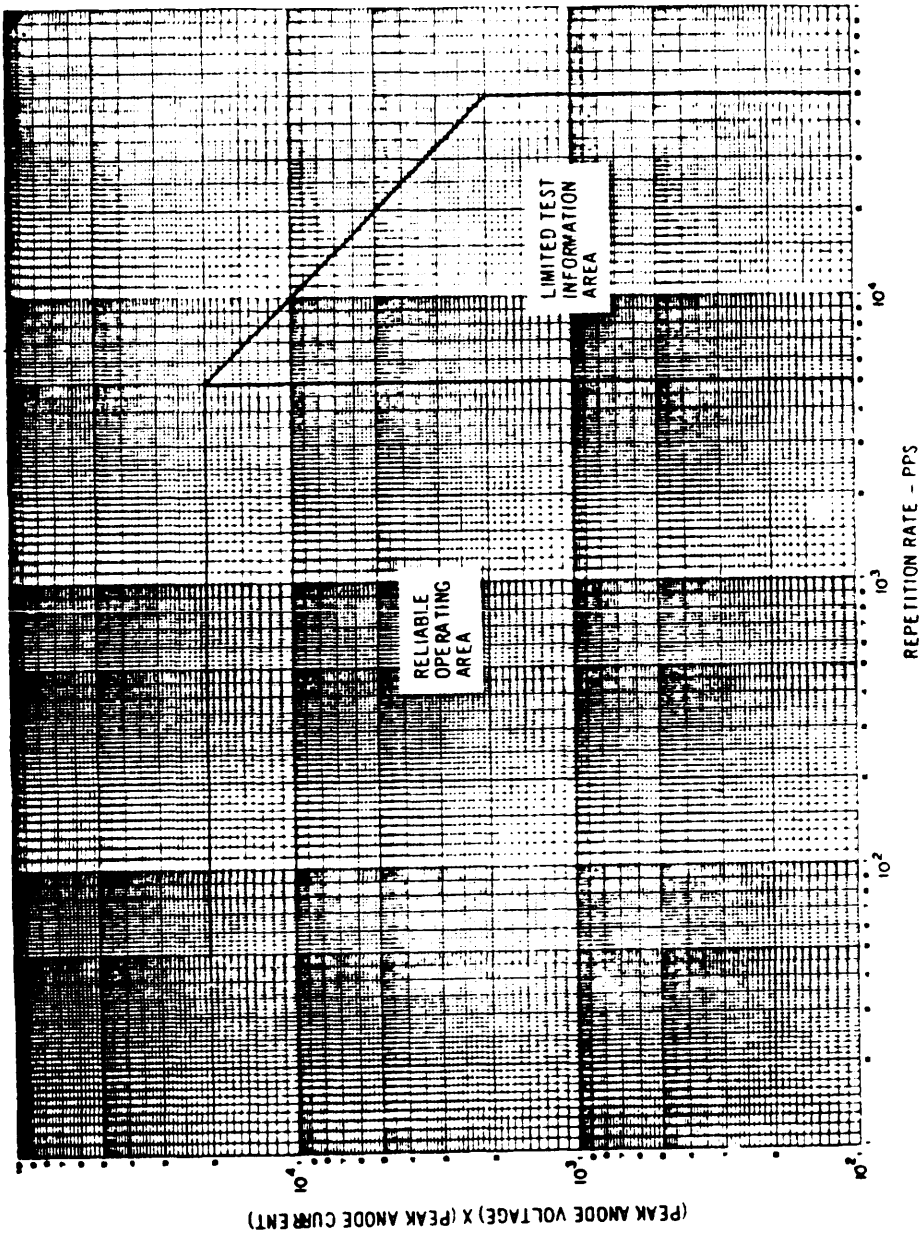



FIGURE 2. Graphical representation of heat factor.



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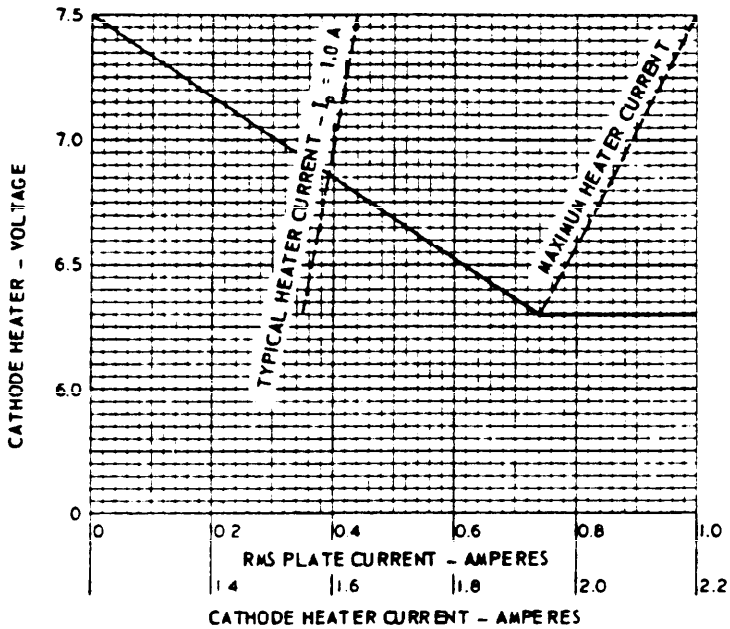



FIGURE 3. Heater voltage versus operating condition.

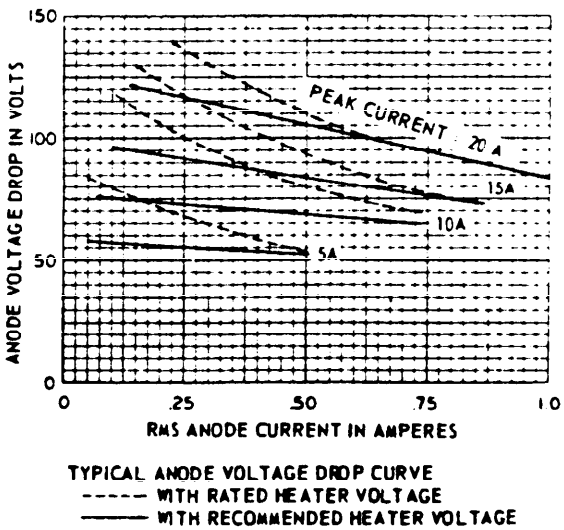



FIGURE 4. Anode voltage versus RMS anode current curve.



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